Yutian Yang

501-368-9640 | charlieyang990314@gmail.com | charlieyang1557.github.io/aboutme/ | linkedin.com/in/yutianyang

Education

University of Illinois Urbana-Champaign • Master of Computer Science — 08/2026 – Expected 12/2027

University of California, Davis • *Master of Science* • Statistics — 09/2021 – 06/2023

University of California, Davis • *Bachelor of Science* • Statistics & Economics — 09/2017 – 06/2021

Skills & Tools

- **Programming:** SQL, Python, R
- **Machine Learning:** Scikit-learn, PyTorch, Random Forests, SVM, CNN, Prophet, Statsmodels
- **AI / LLM:** RAG, Hybrid Retrieval, Vector Search, LLM Evaluation, Prompt Engineering, Anthropic Claude, AWS Bedrock
- **Data Engineering:** Pinecone, FAISS, BigQuery, DBT, AWS, GCP, ETL, Schema Design
- **BI & Analytics:** Tableau, Power BI, Looker, Sigma, Google Analytics

Work Experience

Onshape by PTC (Cloud-Based CAD Platform) – Data Science Intern — 06/2025 – 08/2025
Boston, MA

- Built and evaluated multiple unsupervised anomaly detection models (Prophet + Isolation Forest, Merlion, LSTM-AE) for API telemetry monitoring; selected and deployed Prophet-based pipeline for best performance, interpretability, and ease of deployment.
- Applied AWS Bedrock's Titan Embeddings and Claude 3.5 Sonnet to cluster and analyze NPS feedback, surfacing **top customer concern themes** that directly informed targeted product strategies to improve resolution efficiency.
- Conducted keyword analysis on AI Advisor queries, identifying **~8% overall coverage gaps** with **5% of top topics** lacking sufficient resolution content; delivered prioritized gap findings to guide content improvements.
- Developed Looker dashboards and automated Slack alerts surfacing top 20 daily API anomalies, reducing engineer triage from full manual review to **5 flagged cases** with anomaly metrics pre-computed, significantly cutting inspection overhead.

Pinecone (AI Vector Database Unicorn) – Data Science Intern — 06/2024 – 08/2024
New York, NY

- Designed and built the Book of Business and Account 360 dashboards using SQL and Sigma, **improving sales operations by 15%**. Implemented Row-Level Security (RLS) for tailored views, and documented processes in Notion, **reducing onboarding time by 30%** and ensuring consistent use across teams.
- Developed the "dim_assistants" schema and implemented it in the pipeline using BigQuery and DBT. Created the Pinecone Assistant dashboard using SQL and Sigma, enabling comprehensive tracking of assistant metrics and increasing product team report frequency and cross-functional visibility into assistant performance.
- Conducted churn analysis using Python and Random Forests on upstream unstructured data with missing and incomplete records; applied time series cross-validation to reduce overfitting and maximize effective sample size, identifying **5 key churn metrics** and setting up alerts, **reducing churn by 10%**.

Allschool (EdTech Platform) – Data Analyst Intern — 06/2022 – 08/2022
San Mateo, CA

- Enhanced impression targeting strategies through **A/B testing and segmentation analysis** of user traffic and revenue across regional and platform data using Google Analytics, directly informing channel prioritization decisions.
- Evaluated user behavior across multiple advertising channels using SQL and BI tools, leading to a **15% reduction in project budget** and measurable growth in daily active users.
- Designed a real-time web scraper with Python and Selenium, **accelerating the class selection process by 50%**.
- Developed a key metrics dashboard for active users, daily traffic, and revenue, improving business visibility and supporting data-driven decision making using Google Looker Studio.

UC Davis Department of Economics (Data Analysis & Modeling) – Research Assistant — 07/2020 – 09/2020
Davis, CA

- Analyzed behavioral trends in procrastination and present-biased behavior using a generalized linear model (GLM) and Logistic Regression in a study with Professor Anujit Chakraborty.
- Enhanced the reliability of study results by employing Bootstrapping resampling techniques to **expand the sample size to approximately 20,000 data points**.
- Addressed multicollinearity among predictors and **improved prediction accuracy** of procrastination behavior variables with regularization using Lasso Regression.
- Uncovered procrastination behavior patterns and present-bias signals, translating findings into actionable insights for designing user-centric technology products.

Projects

BoardGame Rules Oracle [Live Demo] — Python, React, FastAPI, Pinecone, Anthropic Claude API

- Built a multi-hop RAG Q&A system for board game rules with **hybrid retrieval** (Pinecone dense + BM25 sparse + RRF fusion), a cross-encoder reranker, and a **3-tier router** (direct answer / iterative multi-hop / honest uncertainty fallback).
- Implemented **citation verification** via string overlap + LLM entailment, achieving **91% accuracy** on Splendor/Catan and **90% Tier-2 multi-hop accuracy** on Food Chain Magnate (vs. 65% target); deployed full-stack on Railway with Docker.

Resume RAG Chatbot [Live Demo] — Python, React, FastAPI, FAISS, Anthropic Claude API

- Built a RAG chatbot with **hybrid retrieval** (BM25 + FAISS), Claude LLM streaming, and query rewriting. Alpha sweep via LLM-as-judge found **pure semantic retrieval achieved 5.0/5.0 relevance**.
- Deployed full-stack app (React + FastAPI) with source attribution, dark/light mode, rate limiting, and **Docker containerized** one-command startup.